

Meridian Aerosystems, Inc.

Response to RFI FA8629-26-RFI-4ISR — Attributable ISR Aircraft
Submitted to: AFLCMC/WI CDO, Wright-Patterson AFB
Date: 12 May 2026
POC: Dr. Elaine Vasquez, VP Government Programs — evasquez@meridianaero.example — (937) 555-0141
CAGE: 7XK42 | Proposed platform: MA-240 "Kestrel" Group 3 UAS
Distribution: Proprietary. Contains competition-sensitive information.

1. Executive Summary

Meridian Aerosystems proposes the MA-240 Kestrel, a production Group 3 attributable ISR platform with 61 airframes delivered to date. The Kestrel meets or exceeds every threshold requirement in the RFI and achieves the majority of objective values in its current production configuration. Responses below follow the RFI KPP/KSA table structure.

2. KPP Compliance — Aircraft (KPP ID 1)

Req ID	Requirement	Vendor Response
1.1	Range	1,650 km from L/R location to collection area (exceeds objective of 1,500 km). Demonstrated during Yuma trials, March 2026.
1.2	Loiter Time	22 hours on station at 10,000 ft MSL, excluding ingress/egress (exceeds 20 hr objective).
1.3	Cruise Altitude	Operational ceiling 24,000 ft MSL; typical ISR cruise 12,000-18,000 ft MSL.
1.4	Climb Rate	1,150 ft/min at max gross weight, sea level standard day.
1.5	Max Speed	128 KTAS dash; 74 KTAS loiter speed.
1.6	Operational Environment	Certified for operations in rain up to 25 mm/hr, winds to 30 kt sustained / 40 kt gusts, temperatures -34C to +49C. Full environmental qualification report available on request.
1.7	Operational Attributes	Two-person launch crew; system deploys from two standard transit cases; runway independent (pneumatic catapult launch, net or skid recovery). Setup to first launch: 45 minutes. Meets T=O.
1.8	Integration	Compliant (T=O). Kestrel provides 800 W continuous / 1.2 kW peak payload power at 28 VDC with MIL-STD-1275 conditioning; integrates sensors, autopilot, communications, and navigation per MOSA-aligned open interfaces.
1.9	Communications	Meets objective: secure BLOS via commercial SATCOM (Ku) with FMV and metadata; Link-16 via integrated STT terminal (fielded 2025).
1.10	Navigation	Meets objective: GPS (SAASM-ready), tactical-grade INS, and alternative PNT via vision-aided navigation in GPS-denied conditions.
1.11	Command and Control	Compliant (T=O): secure BLOS C2 plus fully autonomous pre-programmed mission execution with lost-link contingency profiles.
1.12	Mission Planning	Meets objective: automated route planning, in-flight dynamic re-tasking, sensor scheduling, and air-traffic deconfliction via integrated ADS-B In.
1.12	On-board Processing	Meets objective: edge GPU module supports multi-INT fusion and automatic target recognition (ATR) with operator-in-the-loop confirmation.
1.13	Ground Control Station	Threshold met with objective features: ruggedized laptop GCS (11 kg total footprint) with mission-essential processing; operates from location separate from L/R site. Objective capability provided via optional Meridian Nexus suite: collaborative multi-analyst environment and automated report generation/dissemination to DCGS.

3. KPP Compliance — Subsystems (KPP IDs 2-5)

Req ID	Requirement	Vendor Response
2.1	Communication	Objective met: secure BLOS with FMV and metadata at 52 Mbps sustained; Link-16 integrated.
2.2	Comms Integration	Objective met: integrates with aircraft, sensors, and secure networks (NSA Type 1 encryptor option).
3.1	Navigation	Objective met: GPS + INS + Alt PNT (vision-aided and signals-of-opportunity modes).
3.2	Nav Integration	Objective met: integrated with flight controls and all sensor payloads for geo-registration.
4.1	C2	Compliant (T=O): secure BLOS and autonomous pre-programmed mission modes.
4.2	C2 Integration	Objective met: integrated with flight controls, sensors, and secure networks.
5.1	Onboard Processing	Objective met: multi-INT fusion, ATR, edge processing on 40 TOPS GPU module.
5.2	Processing Integration	Objective met: integrates with all sensors and secure networks.

4. KPP Compliance — Sensors (KPP IDs 6-9)

Req ID	Requirement	Vendor Response
6.1	Image Type	Objective met: multi-spectral (VNIR/SWIR) HD imager. GSD: 8 cm at 12,000 ft AGL nadir.
6.2	Image Storage	Compliant (T=O): 4 TB onboard solid-state storage retains all mission imagery until EUD transfer confirmed.
6.3	Image Integration	Compliant (T=O): in-mission and post-mission download to EUD (ATAK-compatible).
7.1	FMV Field of View	Objective met: gimbaled 360-degree turret plus wide-area persistent surveillance mode (city-scale WAMI at 2 Hz).
7.2	FMV Video Type	Objective met: EO/IR multi-spectral HD (1080p). GSD 12 cm at 10,000 ft slant range 4 km.
7.3	Video Processing Speed	50 Mbps compliant (T=O); demonstrated 55 Mbps sustained.
7.4	MTI Target Tracking	Objective met: single-sensor moving target tracking, vehicle-class targets to 60 km/h.
7.5	MTI Track Capacity	Objective met: 24 simultaneous tracks per sensor.
7.6	Laser Range Finder	Objective met: eye-safe LRF, 10 km range, +/-2 m accuracy.
7.7	Laser Target Indicator	Objective met: NATO STANAG 3733 compatible laser target indication.
7.8	FMV Integration	Compliant (T=O): integrated with aircraft, C3, navigation, and EUD.
8.1	Signal Detector Waveform	Compliant (T=O): SDR-based detector, 30 MHz - 6 GHz, waveform library including push-to-talk, LTE, and PMR; details in classified annex available under appropriate handling.
8.2	Signal Integration	Compliant (T=O): integrated with aircraft, C3, navigation, EUD.
9.1	Other Sensors	Available options: AIS receiver maritime payload and hyperspectral imager (TRL 7).
9.2	Other Sensor Integration	Compliant (T=O) for all offered payloads.

5. Ground Control Station (KPP ID 10) and KSAs

Req ID	Requirement	Vendor Response
10.1	GCS	Threshold met (minimal footprint, mission-essential processing); objective met with Nexus suite (collaborative analysis, automated reporting/dissemination).
10.2	GCS Integration	Compliant (T=O): integrates with aircraft, sensors, C3, navigation, and EUD.

Req ID	Requirement	Vendor Response
KSA-1	Airworthiness	Compliant: MA-240 holds a USAF Group 3 airworthiness assessment (MIL-HDBK-516C basis), issued Nov 2025.
KSA-2	IFF	Compliant: IFF Mode 3/A transponder standard; Mode 5 growth option.
KSA-3	Security	Compliant: authorized to operate on SIPRNet-connected ground segments; IL5 cloud data path.
KSA-4	Manufacturing	Current capacity 14 aircraft/month, scalable to 30/month within 4 months of contract start. First article delivery: 60 days ARO.

6. Sustainment Summary

Meridian offers a holistic maintenance concept with regional Field Support Representatives, spares kits sized for 90-day austere deployments, full technical orders, SCRM program with domestic-source critical components, and operator/maintainer training (10-day course). Unit fly-away cost is provided in the pricing annex.